

Yousef Elrefaey

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WORK EXPERIENCE

Hirth Engines

Stuttgart, Germany

R&D Junior Engineer (*Intern*)

Mar 2025 - Aug 2025

Developed an Excel framework for RTCA DO-160G compliance, managing 23 aviation-standard tests to ensure FAA adherence

- Performed **mechanical and electrical test procedures** on a 2-stroke engine test bench, executing calibration, data logging, and post-test analysis. Reported findings to R&D and test teams, clarifying anomalies and supporting faster decisions. Created and modeled mechanical components in CATIA V5, followed by detailed 2D technical documentation

- Performed precise measurements and created accurate CAD models and drawings from real parts.

- Evaluated **CNC and engine assembly** workflows to improve efficiency and understand precision manufacturing practices.

Observed 2-stroke engine testing procedures—covering **throttle response and electrical validation**—to understand pre-test setups and test room protocols.

Explored and self-navigated 3DEXPERIENCE (Dassault Systèmes) software, strengthening digital engineering and design skills.

EDUCATION

German International University

Cairo, Egypt

Bachelor of Science in Engineering (Automotive Mechatronics)

ABC private School

Abu Dhabi, UAE

A-Levels-British High school

PROJECT EXPERIENCE

German International University

Cairo

Obstacle Avoidance for AV Using ML (*Bachelor Thesis*)

Built and tested a fuzzy logic controller for obstacle avoidance, allowing the robot to navigate autonomously in real-time simulations.

Developed a real-time MATLAB–CoppeliaSim link to control robot actuators and visualize live simulation data

Developed an obstacle detection system with LiDAR and ultrasonic sensors for autonomous wheel control and collision avoidance.

- Implemented path planning in CoppeliaSim to generate and visualize robot trajectories in MATLAB.
- Utilized RL Simulink libraries to model and simulate LiDAR sensor operations.
- Developed a front-drive four-wheel robot with ultrasonic sensors to validate the obstacle avoidance algorithm in real-world tests.

Automatic Car Plate Detection (*Signal and Image Processing Project*)

- Developed a Python-based vision pipeline with RoboFlow augmentation and custom pre-processing to improve plate and character recognition accuracy.

- Extracted plate numbers and alphabets from the detected car plates.

- Implemented a system that provides a percentage accuracy for the detection of car plates.

- Implemented advanced computer vision algorithms leveraging OpenCV for real-time plate detection & OCR.

Self-Driving Car w/ Lane Detection (*Mechatronics Engineering Project*)

- Built a self-driving car that used real-time sensor data to stay in lane and make autonomous driving decisions

- Implemented ultrasonic sensing for real-time awareness, enabling the Arduino to make autonomous navigation decisions.

- Implemented servo motors to control the motion of each wheel independently based on sensor readings.

Designing a Robot Arm (*Engineering Design*)

Designed a robot arm in SolidWorks, optimizing geometry for strength and accurate stress analysis.

- Performed stress-strain analysis on the CAD model to locate weak points and reinforce structural integrity

SKILLS & INTERESTS

Skills: CAD & Design: CATIA V5, SolidWorks, Siemens NX, AutoCAD, 3DEXPERIENCE (Dassault Systems)

SIMULATION & CONTROL: MATLAB & Simulink, Arduino, Sensor Integration **PROGRAMMING:** C++, Python

Interests: Mechanical Designing, Automotive, Aerodynamic analysis using CFD

VERSION CONTROL & COLLABORATION: GitHub

DEVELOPMENT METHODOLOGY: Agile / Scrum

Soft Skills: Communication, Teamwork, Quick Learner, Adaptability, Analytical thinking

Languages: Arabic (**Native**), English (**Fluent**), German (**B1 – Learning**)

Internship certificate

Hirth Engines GmbH
Max-Eyth-Str. 10
71691 Freiberg a. N.
Germany

31st of August 2025

To whom it may concern

Mr. Youssef Mohammed Elrefaey, born on May 2, 2005, completed a mandatory internship at our company from March 3, to August 31, 2025, as part of his studies.

Hirth Engines is globally recognized for its innovative 2-stroke engines, offering unmatched engineering solutions and industry-leading performance.

The internship position gave Mr. Elrefaey the opportunity to gain an insight into an operational process and the work of a design and development employee. In particular, the trainee had the opportunity to acquire the practical professional knowledge in design and development.

During the internship, Mr. Elrefaey was entrusted with the following tasks:

- Creation of instructions for testing of engines based on the applicable environmental testing standard
- Design of parts for our test benches for adapting the engines
- Creation of drawings for the test bench parts
- Collaboration with production and purchasing

Mr. Elrefaey demonstrated quick comprehension and possesses extensive specialist knowledge acquired during his studies, which he applied in a targeted manner.

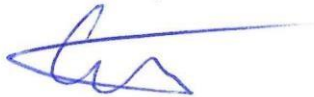
During his internship, he demonstrated his skills with self-motivation to learn new skills.

He carried out his tasks quickly and efficiently at all times and, always maintained an overview and acted calmly and deliberately.

His superiors and colleagues always appreciated him as a hard-working and friendly intern. His personal conduct was always impeccable.

Mr. Elrefaey is leaving our company after the expiry of the previously agreed period.

We wish him every success in his further professional and personal endeavors.



Torsten Windorfer
Head of Research & Development



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Certificate of Completion

This certifies that **Youssef Mohamed Elrefaey** has successfully completed the Internship program at **Hirth Engines GmbH, Max-Eyth-Straße 10, 71726 Benningen, Germany** from **10.03.2025 to 25.08.2025**.

During this period, Youssef Mohamed Elrefaey has actively contributed to the assigned tasks, demonstrated commitment and professionalism, and successfully fulfilled all requirements of the internship.

We hereby recognize and appreciate the achievements and dedication shown throughout the internship.

Issued on **01.09.2025**



Internship Mentor

Torsten Windorfer

CTO



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 Organization Representative

Henning Nitzsche

Site Manager / Head of Supply Chain &
Administration / POA